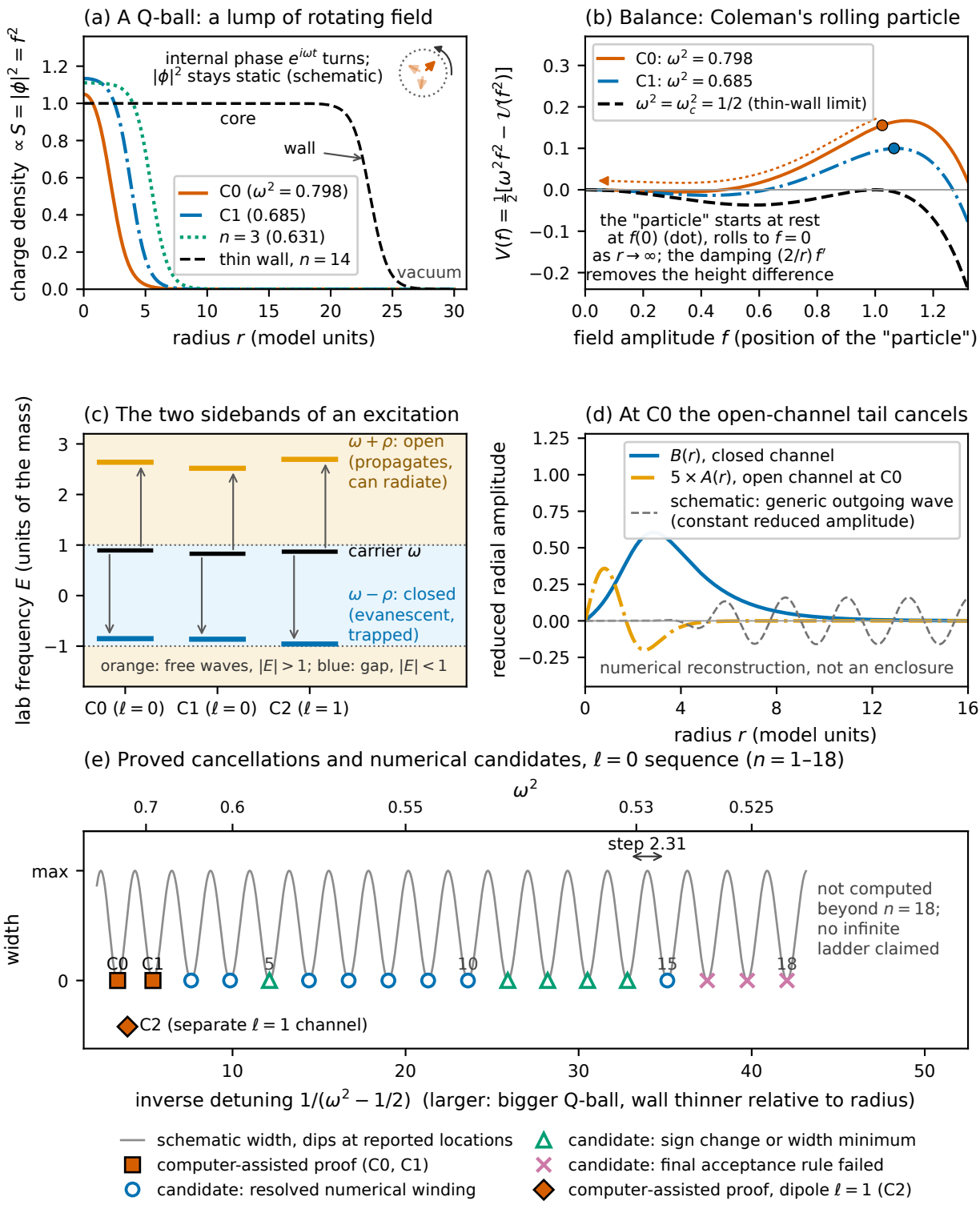

(a) A Q-ball: a lump of rotating field
internal phase $e^{i\omega t}$ turns;
$|\phi|^2$ stays static (schematic)
core
wall
vacuum
C0 ($\omega^2 = 0.798$)
C1 (0.685)
$n = 3$ (0.631)
thin wall, $n = 14$
charge density $\propto S = |\phi|^2 = f^2$
radius r (model units)
(b) Balance: Coleman's rolling particle
C0: $\omega^2 = 0.798$
C1: $\omega^2 = 0.685$
$\omega^2 = \omega_c^2 = 1/2$ (thin-wall limit)
the "particle" starts at rest
at $f(0)$ (dot), rolls to $f = 0$
as $r \to \infty$; the damping $(2/r)\,f'$
removes the height difference
$V(f) = \frac{1}{2}[\omega^2 f^2 - U(f^2)]$
field amplitude f (position of the "particle")
(c) The two sidebands of an excitation
$\omega + \rho$: open
(propagates,
can radiate)
carrier ω
$\omega - \rho$: closed
(evanescent,
trapped)
orange: free waves, $|E| > 1$; blue: gap, $|E| < 1$
C0 ($\ell = 0$) C1 ($\ell = 0$) C2 ($\ell = 1$)
lab frequency E (units of the mass)
(d) At C0 the open-channel tail cancels
$B(r)$, closed channel
$5 \times A(r)$, open channel at C0
schematic: generic outgoing wave
(constant reduced amplitude)
numerical reconstruction, not an enclosure
reduced radial amplitude
radius r (model units)
(e) Proved cancellations and numerical candidates, $\ell = 0$ sequence ($n = 1$–18)
ω^2
step 2.31
not computed
beyond $n = 18$;
no infinite
ladder claimed
C2 (separate $\ell = 1$ channel)
width
inverse detuning $1/(\omega^2 - 1/2)$ (larger: bigger Q-ball, wall thinner relative to radius)
schematic width, dips at reported locations
computer-assisted proof (C0, C1)
candidate: resolved numerical winding
candidate: sign change or width minimum
candidate: final acceptance rule failed
computer-assisted proof, dipole $\ell = 1$ (C2)